

Lecture 3: <Convexity>

1 Optimization and Convexity

A general optimization problem seeks a decision x that minimizes an objective function:

$$\min_{x \in \mathbb{R}^n} f(x).$$

In a general nonconvex problem, the objective may have several local minima in addition to a global minimum. Convexity gives a useful structure under which local and global optimality coincide.

Definition 1 (Convex function). *A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex if, for all x, y and every $\alpha \in [0, 1]$,*

$$f(\alpha x + (1 - \alpha)y) \leq \alpha f(x) + (1 - \alpha)f(y).$$

If the inequality is strict whenever $x \neq y$ and $\alpha \in (0, 1)$, then f is strictly convex.

Definition 2 (Convex optimization). *A convex optimization problem has the form*

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & f(x) \\ \text{subject to} \quad & x \in X, \end{aligned}$$

*where f is convex and X is a convex set. The central property is that **any local minimizer is also globally optimal**.*

2 Characterizations and Consequences

For a differentiable function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, convexity can be characterized using derivatives.

2.1 First-order characterization

A differentiable convex function satisfies

$$f(y) \geq f(x) + \nabla f(x)^\top (y - x), \quad \forall x, y.$$

Thus, the first-order approximation at any point x is a global lower bound on the function.

2.2 Second-order characterization

For a twice-differentiable function, convexity is characterized by a positive semidefinite Hessian:

$$\nabla^2 f(x) \succeq 0, \quad \forall x.$$

Two useful consequences are:

- The set of global minimizers of a convex objective is convex.
- A strictly convex objective has at most one minimizer.

3 Quadratic Example

Consider

$$f(x, y) = \frac{1}{2} (\alpha x^2 + \beta y^2) - x.$$

Its Hessian is

$$\nabla^2 f = \begin{bmatrix} \alpha & 0 \\ 0 & \beta \end{bmatrix}.$$

The signs of α and β determine the curvature:

- $\alpha = 1, \beta = 0$: convex, but not strictly convex.
- $\alpha = 1, \beta = 1$: strictly convex.
- $\alpha = 1, \beta = -1$: nonconvex.

4 Linear Regression as a Convex Problem

The lecture uses restaurant ratings as an example. Let x_i denote the rating features for restaurant i , and let y_i denote Chloe's rating. A linear predictor models the rating as

$$y_i \approx \theta^\top x_i,$$

with squared loss

$$(\theta^\top x_i - y_i)^2.$$

Stacking the observations gives

$$X = \begin{bmatrix} x_1^\top \\ \vdots \\ x_n^\top \end{bmatrix}, \quad y = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}.$$

The linear regression problem is

$$\min_{\theta} \|X\theta - y\|^2, \quad \theta \in \mathbb{R}^n.$$

For the scaling used in the lecture, the derivatives are

$$\nabla f(\theta) = X^\top (X\theta - y), \quad \nabla^2 f(\theta) = X^\top X \succeq 0.$$

Therefore, the least-squares objective is convex. Setting the gradient equal to zero gives the normal equation

$$X^\top X \theta^* = X^\top y.$$

A solution exists, and the solution is unique when X has full column rank.

5 Sparse Linear Regression and Convex Relaxation

For a more interpretable predictor, we may want many entries of θ to be zero. A direct sparsity formulation is

$$\min_{\theta} \|X\theta - y\|^2 + \lambda \|\theta\|_0, \quad \theta \in \mathbb{R}^n,$$

where $\|\theta\|_0$ counts the number of nonzero entries of θ . This objective is nonconvex and discontinuous.

A convex relaxation replaces the ℓ_0 penalty with the ℓ_1 norm, giving LASSO:

$$\min_{\theta} \|X\theta - y\|^2 + \lambda \|\theta\|_1, \quad \theta \in \mathbb{R}^n.$$

The ℓ_1 penalty encourages entries of θ to become exactly zero. The resulting problem is convex and efficiently solvable, although it is nonsmooth.

Key idea. Replace a combinatorial penalty by a convex surrogate.